# Electricity Systems Design for Standalone, Minigrid, Grid-Connected Configurations with Domestic and Flexible Irrigation Loads: A Case Study in Tigray, Ethiopia

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***Abstract*—Irrigation is critical for increasing food production in Sub-Saharan Africa (SSA), but the lack of electricity in rural areas is a significant barrier. This paper investigates electricity system solutions for a cluster of irrigated smallholder farms in Tigray, Ethiopia, detected by a previous advanced model. These areas have suitable conditions for irrigation and clear existing energy demands, making electricity systems likely to succeed. Irrigation loads have flexibility, allowing them to align with solar energy availability and utilize soil water storage over several days. However, irrigation seasonality and the minimum pump power requirements potentially reduce electricity system utilization rates. We used an existing network design model and developed a generation system design model to examine standalone, minigrid, and grid-connected systems. Results show that shared solar systems and the flexibility of irrigation loads are crucial in maintaining low electricity costs. Small-scale minigrids connecting nearby farms ensure efficient system utilization and avoid the high costs associated with constructing regional networks using medium voltage wires and transformers, enhancing their viability under current conditions. These findings can guide infrastructure investments for rural electrification with irrigation loads in SSA.**

***Index Terms*—irrigation, flexibility, electrification, solar**

## I. Introduction

Increasing food production in Sub-Saharan Africa (SSA) is essential for supporting the growing population and boosting farmers' economic welfare. The region's average cropping intensity—the number of crops grown within a single agricultural year—is 1.29. This number could rise to 2 under local temperature conditions, or to 1.5 when considering both temperature and precipitation [1]. Irrigation is critical for increasing rainfed agricultural production [2] and allows for cultivation during dry seasons. However, only 4% to 6% of the land is irrigated in SSA, compared to over one-third in Asia [3].

Modern irrigation links closely with energy use [4]. For example, a single crop of maize typically requires 7,266 cubic meters of water per hectare [5], which may translate to over a thousand kWh of electricity during the dry season. Farmers in SSA often lack access to affordable modern energy, especially in rural areas where grid connections are absent [6]. Thus, the lack of energy is a significant barrier for farmers adopting irrigation. When irrigation requires affordable electricity, local electrification also benefits from irrigation, as they need productive uses of energy to serve as anchor loads—a role that irrigation can effectively fulfill [7]. Consequently, public and private sectors are seeking locations with irrigation demands to determine where to invest in electricity infrastructure.

The presence of irrigated smallholder farms in rural SSA indicates that farmers have met other essential requirements, such as access to suitable soil, seeds, fertilizer, water sources, and markets [8]. In these areas, energy demands are evident. Currently, a significant portion of these demands is met through manual labor, with motorized pumps used to a lesser extent [9], [10]. This implies the potential benefits of cost-effective electricity infrastructure. In a conducted shared-solar irrigation project in Senegal by one of our co-authors [11], farmers benefited from shared-solar systems for crop irrigation and were willing to use such systems. With this positive signal, this paper further investigates solar energy for irrigation loads.

The co-authors and colleagues of this paper have developed a method for detecting irrigation on smallholder farms [12]. It is achieved by identifying farms actively cropping during dry seasons, with results at a 10-meter resolution in Ethiopia's Tigray and Amhara regions. The method begins with an analysis of MODIS imagery using the Enhanced Vegetation Index time series to identify seasonal rainfall patterns and guide labeling. It then uses Sentinel-2 imagery to analyze labeled pixels that show dry season greening and senescence cycle. Additionally, field-collected irrigation data provide the ground truth. All these data were used to train classifiers for detecting irrigation. Results show a transformer-based neural network architecture achieves over 95% accuracy in distinguishing irrigated from non-irrigated cropping lands.

This paper uses the results from this irrigated land detection study [12]. As a case study, we selected a region in Tigray, Ethiopia, spanning 5 to 6 kilometers and encompassing a total of 125 hectares of smallholder irrigated lands. We then explore different electricity system solutions in this region, with findings applicable to similar regions.

The electricity system solutions explored in this study encompass stand-alone systems and minigrid systems. For minigrids, considerations include adopting a single system that covers the entire region or multiple smaller systems. Additionally, the grid connection solution is also briefly assessed. On the demand side, the analysis includes irrigation and domestic loads. On the generation side of the stand-alone or minigrid systems, solar, battery, and diesel options are considered, as they are widely feasible and popular in many rural areas. An existing electricity network design model [13] and a newly developed generation system design model for this study are applied to simulate these different solutions.

Since irrigation represents the major demands, this paper wants to highlight three key characteristics of irrigation loads: flexibility, seasonality, and the minimum power threshold.

**Flexibility**: Irrigation scheduling can leverage natural conditions. The soil can store water from irrigation and rainfall, thereby sustaining crops for several days, so irrigation loads can exhibit flexibility over a period of days. Moreover, irrigation activities can be timed to align with the availability of solar energy, allowing the daily flexibility of irrigation loads.

**Seasonality**: In the land identified as irrigated, there are two cropping seasons: one in the rainy season and the other in the dry season, followed by fallow periods. The dry cropping season has the highest irrigation loads, the rainy season has less due to rainfall, and the fallow season has none. This variation in irrigation demand adds complexity to the design of electricity systems. On the other hand, during the dry season, solar panels can typically provide consistently high outputs.

**Minimum power threshold**: Driving a pump typically requires a minimum power threshold, posing a challenge for smallholder farmers designing stand-alone electricity systems. From an energy perspective, small-capacity solar panels might suffice, but they are insufficient to operate a pump. However, deploying enough solar panels to drive pumps results in a significant energy surplus, leading to waste [11].

Given these characteristics, a practical solution involves shared generation sets and managed irrigation scheduling, which can meet pump power capacities and exploit the flexibility of irrigation loads. There is evidence that farmers are open to cheaper electricity, agree to irrigation scheduling, and are willing to share energy systems with others [8]. However, implementing a sharing system requires connection infrastructure with additional costs. This underscores the importance of conducting this study to explore effective solutions.

The innovations of this paper include highlighting the importance of irrigation load flexibility and introducing an electricity system design model that effectively accommodates both flexible irrigation and domestic loads. Additionally, the paper compares standalone, minigrid, and grid-connected electricity solutions to determine their suitability.

## II. Methodology

### A. Configurations and Models Overview

Before introducing the configurations, it is essential to define the irrigation zones in this study. In the case study areas, 10-meter resolution irrigation prediction pixels are aggregated into polygons. These polygons are divided into circles into multiple 300-meter radius circles by placing centers at regular intervals. It then optimizes the placement of these circles to maximize their overlap with the irrigation areas. These optimized circles are designated as the irrigation zones. Each zone may contain multiple smallholder farms with domestic and irrigation demands. The 300-meter radius was chosen based on field lessons suggesting it is practical. A verification calculation shows that a 3 kW pump with a 50mm² conductor experiences about a 5% voltage drop over 800 meters [14]. In the network design used in this paper, the distance from the transformer to the circle centroid is limited to 500 meters, making a 300-meter extension practical for irrigation zones.

This study examines five electricity system configurations. Configurations 0 and 1 are standalone systems. Configurations 2 and 3 are minigrids of different scales. Configuration 4 is a grid-connected system with the national electricity supply.

- **Configuration 0:** Each farm has an independent generation system for domestic loads, with no connection wires.
- **Configuration 1:** Each farm has an independent generation system for domestic and irrigation loads, with no connection wires.
- **Configuration 2:** Each irrigation zone has a generation system shared by farms inside the zone, supplying the domestic and irrigation loads. This requires low-voltage (LV) wires to connect farms in each zone.
- **Configuration 3:** A single generation system serves the entire region, supplying the domestic and irrigation loads. This requires medium-voltage (MV) and LV wires to connect farms and irrigation zones.
- **Configuration 4:** The region has access to grid supply electricity at $0.06/kWh, but requires MV and LV wires to connect farms and irrigation zones, meeting domestic and irrigation loads.

Fig. 1 presents the energy system design process flowchart for Configuration 3, which includes electricity generation and connection design models. Other configurations adopt parts of this flowchart. Configurations 0 and 1 implement the generation system design model for each farm but do not use the network design model. Configuration 2 uses the generation system design model for each irrigation zone and includes intra-zonal LV wires but omits the network design model. Configuration 4 employs the network design model and considers intra-zonal LV wires but does not use the generation system design model.

Simone Fobi and colleagues developed the two-level network design algorithm [13]. This algorithm inputs node locations and optimizes total connection costs to determine optimal transformer placements and LV and MV networks to connect all nodes. In Configurations 3 and 4, the centroids of irrigation

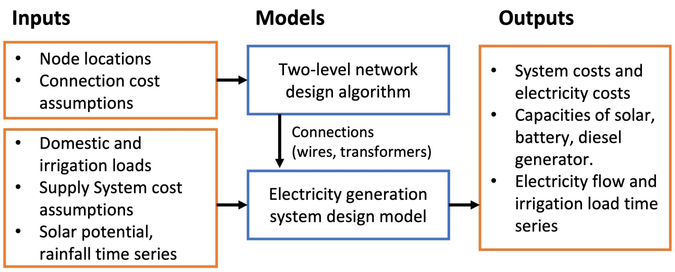


Fig. 1. Energy system design flowcharts with models and their inputs/outputs

zone circles are used as the node locations input for the algorithm. Connection cost assumptions are detailed in Table II. Additionally, in Configurations 2-4, within each irrigation zone, the LV wiring required to connect farms is calculated at 150 meters per hectare of irrigated land.

The electricity generation system design model is employed in Configurations 0-3. This model aims to find the least cost solution to meet electricity loads. It outputs optimal combinations of solar, battery, and diesel generators and associated costs. To address computational issues associated with integer variables from irrigation minimum power constraint, a two-phase structure is introduced, as described in section II-C. The model is coded in Python and uses the Gurobi solver.[1]

### *B. Model Inputs and Assumptions*

The input domestic load is a fixed annual hourly time series, established through two steps. Firstly, the average diurnal load shape of customers is derived from pre-existing laboratory data. Secondly, this load shape is scaled to match the average daily demand of each modeling scenario. This profile is replicated daily to complete the series for the entire year. The domestic loads show an obvious evening peak and remain relatively constant and low at other times. The complete hourly loads are available in the repository mentioned.

Irrigation loads have flexibility over days due to soil water storage capabilities. These loads are determined by the average daily water requirement for crops and the cropping land area. The model schedules irrigation loads and considers rainfall to ensure water needs are met within any continuous five-day period during any cropping season.

In the case study, two cropping periods require irrigation annually. The first is in the rainy season from July 1st to October 30th; the second is in the dry season from November 15th to March 15th of the following year. The periods outside these dates are considered fallow, with no water requirements. The daily water requirements for crops are estimated at 7 mm, based on general assumptions for typical agricultural practices [5], [15]. To convert water requirements to energy demands, we account for Earth's gravitational pull at 9.8 m/s$^2$, the total water head of 50 m, and the electric pump efficiency of 60%. As a result, irrigating one hectare of land needs approximately 15.88 kWh of electricity per day. This study primarily focuses on the energy system design, employing generalized assumptions regarding water requirements to inform our analysis.

In Configurations 0 and 1, we simulate a typical Tigray farming household with 0.5 hectares of land [16] and a daily domestic energy use of 1 kWh. Configuration 0 accounts for only the domestic load, while Configuration 1 includes domestic and irrigation loads. For these two configurations, the total number of households in the region is determined by dividing the total irrigated land area by 0.5 hectares. The total installed capacities are then calculated by multiplying the number of households by the capacities per household. In Configuration 2, model simulation is done for each irrigation zone. The ratio of the irrigated land area to 0.5 hectares is calculated to scale the domestic load. Both domestic loads and irrigation requirements for each irrigation zone are input to obtain the results. Finally, capacities for each zone are summed. Configuration 3 simulates the entire region's domestic load and irrigation requirement once. The domestic load is scaled based on the ratio of total irrigated land to the land area of a single farmer.

The cost structures and modeling parameters for the electricity generation system design model and two-level network design model are specified in Table I and Table II. The cost figures presented in both tables encompass capital, installation, and ancillary components. However, this paper excludes meter and in-house wiring costs and the relatively low maintenance expenses for solar panels, batteries, and diesel generators.

Particularly, this paper explores the impact of installation scale on solar panel costs. Larger scale installations at a single site generally offer reduced costs, while smaller projects face higher per-unit costs [11]. To quantify these scaling effects, a step-wise cost structure is adopted. The solar system costs are interpolated from the data in Table I. For the endpoints, installations smaller than 5 kW are priced at $1,336 per kW, while projects exceeding 1,000 kW are priced at $573 per kW. These costs are derived from the economies of scale for Photovoltaics (PV) system sizes in the US [17] and are adjusted for Ethiopia by comparing per-kWh costs of distributed solar projects in developing countries with those in the US [18]. To contextualize the scenarios in this paper, a single farmer might require less than 5 kW, an irrigation zone might need 10-50 kW, and regional projects could require several hundred kilowatts.

TABLE I
STEP-WISE COST STRUCTURE OF SOLAR PV SYSTEMS

| **System Size (kW)** | 5 | 10 | 50 | 250 | 1000 |
|---|---|---|---|---|---|
| **Costs ($)** | 6,680 | 11,450 | 44,550 | 183,000 | 573,000 |

The electricity generation system design model incorporates two weather information inputs: rainfall and solar radiation. For this case study, both data are from July 2014 to June 2015. The daily rainfall time series for a site in Tigray was previously collected by our laboratory, which shows a total rainfall of

[1]The model can be found at: https://github.com/SEL-Columbia/generation-design-for-flexible-irrigation.

TABLE II
ENERGY SYSTEM DESIGN PARAMETERS AND COSTS

| Parameter | Unit | Value |
|---|---|---|
| Solar lifespan * | year | 15 |
| Minimum solar capacity to install * | kW | 0.2 |
| Lead-acid battery storage cost a | $/kWh | 181 |
| Lead-acid battery round efficiency b | % | 80 |
| Lead-acid battery max depth of discharge b | % | 60 |
| Lead-acid battery lifespan * | year | 5 |
| Battery inverter cost a | $/kW | 173 |
| Battery inverter lifespan * | year | 10 |
| Diesel generator cost a | $/kW | 808 |
| Diesel fuel price c | $/liter | 1 |
| Diesel fuel energy density d | MJ/liter | 38.6 |
| Diesel fuel efficiency e | % | 30 |
| Diesel generator lifespan * | year | 10 |
| Minimum diesel generator capacity to install * | kW | 0.5 |
| Medium voltage wire cost a | $/m | 16 |
| Low voltage wire cost a | $/m | 8.45 |
| Transformer cost a | each | 4,250 |
| Generation side transformer cost a | each | 10,000 |
| Connection equipment lifespan * | year | 15 |
| Financial discount rate f | % | 12.8 |

a Cost estimates are sourced from [19], including capital and installation costs.
b Battery parameters are sourced from [20].
c Considering the average diesel price was $0.8 per liter between 2006 and 2016 [21], and $1.3 per liter in 2023 [22], this study adopts a medium price of $1 per liter.
d Diesel energy density is sourced from [23].
e The model simplifies the efficiency to a single value, based on [24].
f The average value is used, based on [25].
* These parameters are assumptions validated by experts familiar with Ethiopian rural electricity systems.

683 mm in the modeling period, with a single obvious rainy season from late June to October. The solar time series was obtained from an online tool [26], using the centroid location of the case study area. In the tool, settings were configured as follows: utilizing the SARAH2 database, setting a default loss factor of 14%, and choosing to optimize the slope and azimuth angles for fixed-axis panels. This tool computes the hourly generation potential in kWh per 1-kW solar panel.

### C. Model Formulation

With all the assumptions and inputs established, the energy generation system design model minimizes the annual costs associated with deploying and operating generation technologies, as outlined in (1). These technologies include solar panels, battery storage, battery inverters, and diesel generators. Decision variables $X_{\text{tech}}$ represent the capacity of each technology, measured in kW for solar panels, inverters, and diesel generators, and in kWh for battery storage. Unit costs are expressed as $C_{\text{tech}}$ in \$/kW or \$/kWh. Annualized costs, denoted by $A$ and detailed in (2), are calculated by the lifespan $L$ of each technology and the discount rate $i$. Operational costs account for diesel fuel expenses, computed from consumption ($F_{df,t}$ in liters) and unit price ($P_{df}$ in \$/liter).

$$\text{minimize} \left( \sum_{\text{tech}}^{\text{tech-list}} X_{\text{tech}} C_{\text{tech}} A_{L,i} + \sum_{t} F_{df,t} P_{df} \right) \tag{1}$$

$$A_{L,i} = \frac{i(1+i)^{L_{\text{tech}}}}{(1+i)^{L_{\text{tech}}} - 1} \tag{2}$$

Energy balance is an hourly constraint, ensuring supply equal to demand, as detailed in (3). Demand includes domestic and irrigation loads, denoted by $E_{\text{dome},t}$ and $E_{\text{irri},t}$. While the domestic load for each hour is fixed as input, the irrigation load is treated as a decision variable due to its flexibility. Supply is comprised of utilized electricity from solar, $U_{\text{solar},t}$, and diesel, $U_{\text{diesel},t}$, both constrained by their respective capacities, in (4) and (5). Additionally, solar generation is limited by the hourly potential $W_{\text{solar},t}$ in kWh/kW. In (6), diesel fuel consumption is calculated using a conversion efficiency $\eta_{\text{diesel}}$, which quantifies how much electricity in kWh can be generated per liter of diesel, based on parameters listed in Table II.

$$U_{\text{solar},t} + U_{\text{diesel},t} + \delta_t - \gamma_t = E_{\text{dome},t} + E_{\text{irri},t} \tag{3}$$

$$U_{\text{solar},t} \leq X_{\text{solar}} W_{\text{solar},t} \tag{4}$$

$$U_{\text{diesel},t} \leq X_{\text{diesel}} \tag{5}$$

$$\frac{U_{\text{diesel},t}}{\eta_{\text{diesel}}} = F_{df,t} \tag{6}$$

As detailed in (7), the battery charging ($\delta_t$) and discharging ($\gamma_t$) rates, adjusted for one-way efficiency $\eta_{\text{batt}}$, should be aligned with changes in the battery state of charge (SoC). The initial SoC at the first hour is set equal to the SoC in the last hour to ensure consistency across the simulation period, as recommended by [27]. This condition is specifically applied when $t = 1$, as outlined in (7). Additionally, the SoC must be maintained within operational limits to preserve equipment lifespan, as specified in (8), where the maximum depth of discharge is $\Delta$. Similarly, the battery discharge and charge rates are limited by the battery inverter capacity, as detailed in (9). Meanwhile, the model assumes a four-hour battery system, indicating that the effective battery storage capacity is four times that of the battery inverter capacity.

$$\frac{\delta_t}{\eta_{\text{batt}}} - \eta_{\text{batt}} \gamma_t = \begin{cases} S_{t-1} - S_t, & \text{if } t > 1 \\ S_T - S_t, & \text{if } t = 1 \end{cases} \tag{7}$$

$$X_{\text{batt}}(1 - \Delta) \leq S_t \leq X_{\text{batt}} \tag{8}$$

$$\gamma_t \leq X_{\text{inv}} \quad \text{and} \quad \delta_t \leq X_{\text{inv}} \tag{9}$$

The water requirement for any continuous five-day period is achieved by soil water storage system modeling. Similar to the battery system, the soil can charge water from rainfall or irrigation and discharge it later. At the end of each day, the maximum soil water storage capacity is limited to 28 mm, equivalent to four days' requirements. Consequently, if rainfall or irrigation occurs on a single day, the soil can sustain crops for the next four days by releasing 7 mm daily.

This approach also accommodates any other scenarios. By constructing this soil water storage system and meeting daily water requirements, the model simplifies the formulation and effectively meets any continuous five-day water requirements.

The irrigation constraints apply only during the rainy and dry cropping seasons. First, the daily irrigation water for crops, $I_d$, is derived from the hourly irrigation load, $E_{\text{irri},t}$, in equation (10). The conversion process utilizes an efficiency, $\eta_{\text{irri}}$, to convert electricity usage into the mass of water irrigated. It is set at 4408 kg/kWh, based on the assumptions. This mass is then divided by the land area in square meters, $\Lambda_{\text{farm}}$, to get the irrigation water depth in millimeters.

Equations (11) to (13) model water storage in the soil. The left-hand side of (11), including irrigation, $I_d$, rainfall, $R_d$, alongside soil-stored water discharge, $\alpha_d$, and recharge, $\beta_d$, must meet or exceed the crop's daily water requirement, $Q$, with all numbers in millimeters. The water storage level per day, $G_d$, changes with charge and discharge, as described in (12). The water storage level is limited by the capacity of the soil, $\Theta_{\max}$, as shown in (13). Excess water results in runoff.

In practice, to prevent inefficient charging and discharging during periods when solar energy supply exceeds demand, we introduce minor costs to the battery charge and soil water charges. This approach minimizes unnecessary electricity flow in the model without impacting the final results. Additionally, at the start of each cropping season, the model defaults the soil-stored water level to zero.

$$\frac{\sum_{t=d\times 24}^{d\times 24+23}(E_{\text{irri},t})\eta_{\text{irri}}}{\Lambda_{\text{farm}}} = I_d \tag{10}$$

$$I_d + R_d + \alpha_d - \beta_d \geq Q \tag{11}$$

$$\alpha_d - \beta_d = G_d - G_{d+1} \tag{12}$$

$$G_d \leq \Theta_{\max} \tag{13}$$

The last constraint is applied only in the second phase of the model. Mandated by (14), it requires the irrigation electric load to be at least 0.5 kW to ensure efficient pump operation and realistic modeling. This helps to avoid situations where underpowered solar systems are used to drive the pumps.

$$E_{\text{irri},t} \geq 0.5 \tag{14}$$

This constraint employs binary variables, increasing the computational complexity of the optimization model. To manage this, a two-phase solving structure was developed. The first phase calculates solar and battery capacities, using the objective function in (1) and constraints from (3) to (13). The aim is to determine the optimal capacities for solar and battery. The second phase uses the solar and battery capacity results from the first phase. It applies the same objective function and constraints, with the addition of (14), and retains the flexibility to adjust diesel generator capacities. The second phase mandates that the pump operates with the minimum power and determines the corresponding operational electricity flows.

## III. Results and Discussion

This section presents our study's findings, starting with a geographic visualization of the region in Fig. 2. The green polygons represent the irrigated lands, totaling 125.3 ha. The surrounding light yellow circles with a 300-meter radius are the 36 irrigation zones. Some circles overlap, but the irrigated field is always assigned to only one zone. The total area covered by these irrigation zones amounts to 901.5 ha. The figure also shows the MV and LV connection results from the two-level network design. There are 17 service transformers used at the points where MV is converted into LV or at the centroids of the single-served irrigation zones. This visualization aids in understanding the layout of irrigated lands and the practical implementation of the two-level network design. Exploring the exclusion of some irrigation zones with small irrigated lands located far from the main cluster of circles, which require long MV lines to connect, could be worthwhile. However, this is not the current focus of this paper.

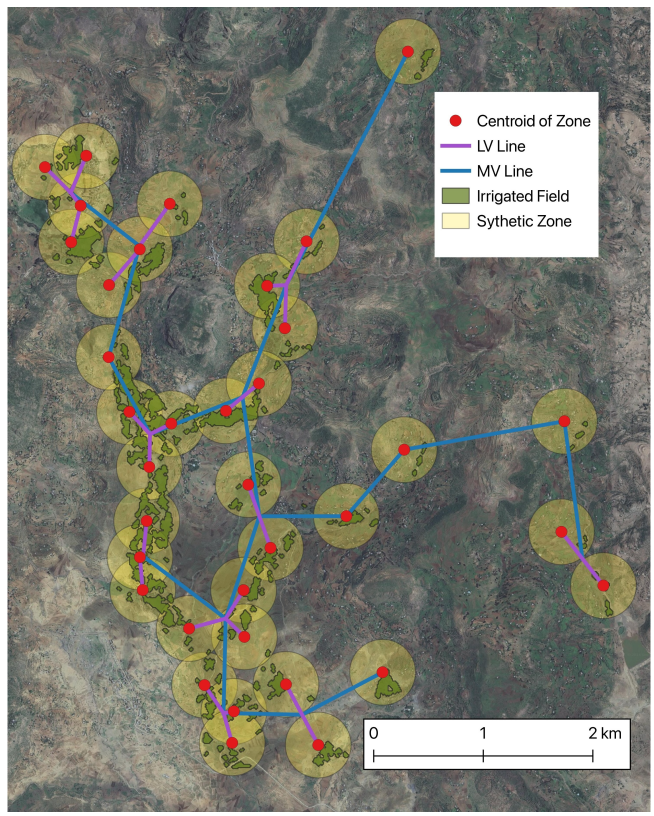


Fig. 2. Irrigated lands layout and modeled medium-voltage (MV) and low-voltage (LV) connections in the case study area in Tigray, Ethiopia. Synthetic 300-meter radius circles represent irrigation zones. Background: Google Satellite Layer (May 2024).

The daily rainfall data and illustrative irrigation requirements are shown in Fig. 3. The figure outlines two cropping seasons: one in the rainy season and another in the dry season. As introduced earlier, the soil has a water storage capacity, while water exceeding this capacity results in runoff. With rainfall variations, irrigation loads are estimated at 838 kWh/ha in the rainy cropping season and 1890 kWh/ha in the dry season, highlighting the significant seasonality. For a household modeled with 0.5 hectares of land, the annual irrigation load is 1364 kWh, and the annual domestic load is 365 kWh. Irrigation loads account for 79% of the total electricity consumption.

This figure also displays solar potential, highlighting differences between rainy and dry cropping seasons. On average, solar panels generate 4.18 kWh/day in the rainy cropping season and 5.59 kWh/day in the dry cropping season for each kilowatt installed. The illustrative irrigation load is for demonstration, but the actual model results will differ due to irrigation flexibility. For example, around 2014-11-15, one day has low solar potential while adjacent days have higher potential. The system could irrigate less on the low solar potential day and more on the surrounding days, so the actual daily irrigation differs from those shown in the figure. In this case, the irrigation flexibility is exerted.

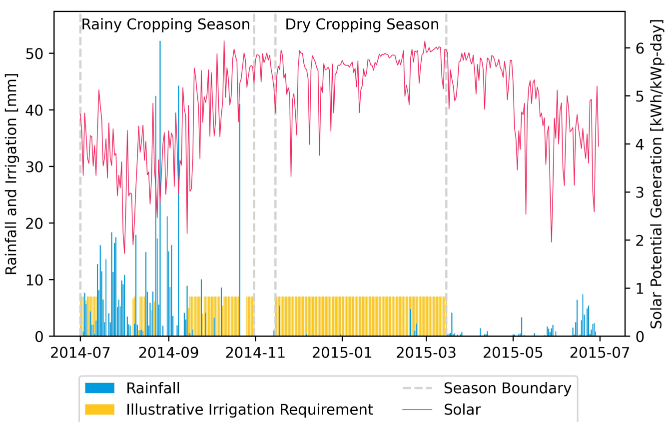


Fig. 3. Rainfall and illustrative irrigation requirements (left y-axis) alongside solar potential (right y-axis), spanning the year with different cropping seasons. The irrigation bars are illustrative and intended to demonstrate potential requirements rather than model outputs.

Table III shows the results of the electricity system design of five configurations, including average daily load, capacities of connection lines and generation systems, and costs. The generation capacities represent the aggregated totals from all systems within each configuration. The final two columns display cost values per kWh of electricity: one for generation costs from solar, battery, and diesel systems; and one for total costs, which include generation as well as connection costs from MV and LV lines and transformers. For configurations with multiple generation systems, the cost values are weighted and averaged.

In Configuration 0, farmers opt for the smallest 0.5 kW diesel generator to meet their domestic loads. Configuration 1 incorporates irrigation loads, prompting a hybrid system with 1.4 kW of solar panels and the smallest diesel generator for each household. Due to varying weather and radiation angles, solar power alone often fails to meet the pump's minimum power threshold, the diesel generator frequently provides supplementary support. Diesel also supplies nighttime domestic loads due to no battery storage in Configuration 1. This highlights the challenge of pure solar systems, as ensuring sufficient solar capacity to drive pumps results in significant surplus solar energy being wasted. Therefore, diesel is always selected for each household. However, by adopting solar power and leveraging the flexibility of irrigation loads over days, Configuration 1 substantially reduces electricity costs from $0.518/kWh in Configuration 0 to $0.270/kWh.

In Configuration 2, farmers within the same irrigation zone are connected and share a generation system. Compared to the previous configurations, diesel generator capacities have decreased because shared solar systems consistently meet the minimum power threshold, eliminating the need for each household to have its own generator. The electricity flow shows that solar predominantly meets the irrigation loads during the day, though diesel occasionally operates to fulfill these demands. The battery size is small, shifting the surplus solar energy to cover part of evening peak loads. The diesel generators still predominantly supply the nighttime loads. The shared system's ample solar power allows for greater flexibility in managing irrigation loads. Compared to Configuration 1, Configuration 2 reduces generation costs, but adding LV lines increases costs by approximately $0.056/kWh, resulting in similar total costs.

Configuration 3 expands into the region-wide connected minigrid, sharing one generation system, which increases the use of solar and battery systems while further reducing diesel generator capacity. Two reasons drive the preference for solar-battery systems over diesel. Firstly, solar energy becomes more cost-effective on a larger scale than multiple smaller systems due to the step-wise cost structures. Despite Configuration 3 having a similar solar capacity to Configuration 2, the solar costs in Configuration 3 are only two-thirds of those in Configuration 2. Secondly, the shared generation system reduces under-utilized diesel generators. From the energy flow perspective, solar primarily meets the irrigation loads during the daytime. At night, diesel primarily supplies the domestic load, while the energy shifted by the battery remains small. In this configuration, there are 17 service transformers and one step-up transformer for the generation system. The infrastructure costs for MV and LV lines and transformers in Configuration 3 exceed $0.20/kWh, making its total costs higher than Configurations 2 and 1. Under the current settings, Configuration 3 is not economically efficient, primarily due to the high expenses associated with the connection infrastructure.

Results from Configurations 2 and 3 indicate that battery capacities are relatively small. In Configuration 3, two subsequent simulations were conducted, excluding either batteries or diesel generators. The results show a 1% cost increase for the solar-diesel system and a 7% increase for the solar-battery system. Thus, batteries and diesel generators can generally replace each other in systems dominated by flexible irrigation loads, though using diesel is slightly cheaper.

Additionally, in Configurations 2 and 3, the utilization rates for solar energy are approximately 57% and 55% respectively, of the maximum potential generation. This is mainly due to irrigation seasonality, as the installed solar energy is wasted

TABLE III
CONFIGURATIONS AND MODELING RESULTS

| Config. | Description | Avg. Daily Demand [kWh/day] | MV line [m] | LV line [m] | Solar [kW] | Diesel [kW] | Battery [kWh] | Generation Costs [$/kWh] | Total Costs [$/kWh] |
|---|---|---|---|---|---|---|---|---|---|
| 0 | Each farm has a generation system for domestic loads, with no connection wires | 253 | 0 | 0 | 0 | 125 | 0 | 0.518 | 0.518 |
| 1 | Each farm has a generation system for domestic and irrigation loads, with no connection wires | 1188 | 0 | 0 | 351 | 125 | 0 | 0.270 | 0.270 |
| 2 | Each irrigation zone has a generation system shared by farms inside the zone, supplying the domestic and irrigation loads. This requires LV wires to connect farms in each zone | 1188 | 0 | 18,795 | 373 | 24 | 66 | 0.206 | 0.262 |
| 3 | A single generation system serves the entire region, supplying the domestic and irrigation loads. This requires MV and LV wires to connect farms and irrigation zones | 1188 | 16,836 | 27,203 | 400 | 16 | 105 | 0.146 | 0.352 |
| 4 | The region has access to grid supply electricity at $0.06/kWh but requires MV and LV wires to connect farms and irrigation zones, meeting domestic and irrigation loads | 1188 | 16,836 | 27,203 | 0 | 0 | 0 | 0.060 | 0.262 |

during fallow seasons and not fully utilized during the rainy cropping season. However, despite significant seasonality, solar remains a cost-effective option for such systems. When the utilization rate is 55% in Configuration 3, the generation cost from solar is about $0.21/kWh.

Configuration 4 demonstrates a national grid-connected scenario, requiring MV and LV lines and 17 transformers. Given the low unit electricity cost of $0.06/kWh, the total costs are $0.262/kWh. Configurations 2 and 4 are the most economical among those presented. Configuration 2 leverages shared generation systems within irrigation zones, eliminating redundant capacities of standalone systems and also avoiding the high costs of larger-scale connections. This aligns with our pilot project in Senegal mentioned above, demonstrating that a shared solar system in a small region is cost-effective. Configuration 4 also performs well when the national grid extension is planned. Meanwhile, these costs in Configuration 4 do not include expenses related to extending the main grid within the region.

This paper further examines the potential for expanded irrigation and greater participation in the energy system if additional local farmers decide to irrigate within the same region. Currently, 125.3 hectares of land is identified as irrigated in the studied region. Considering the irrigation zones total area is 901.5 ha, the irrigated land ratio is approximately 13.9%. Three more synthetic scenarios were simulated, in which the irrigated land and associated domestic load could potentially double, triple, or quadruple.

Fig. 4 displays the results of scaling up irrigation under Configurations 2 and 3, using per-unit total costs as the metric to evaluate changes. Meanwhile, the average daily demand is shown at the top of the figure for reference. The results show that electricity costs decrease as the irrigated land ratio increases for both configurations. Both configurations benefit from slightly improved utilization of the generation system and the economies of scale achieved with larger solar installations, which reduce the capital costs per unit of solar capacity. Additionally, Configuration 3 gains further advantages from the costs associated with connecting zones; increased demand lowers the per-kWh cost due to the amortization of connection investments over a larger volume of electricity demand. Consequently, cost reductions in Configuration 3 are more rapid than in Configuration 2. When irrigated land is tripled, electricity costs for both configurations roughly converge at approximately $0.24/kWh. Further cost reductions are difficult to achieve, with the limit being close to $0.20/kWh due to current solar costs and the seasonality of loads that result in only about half of the solar energy being used.

We also conducted sensitivity tests in four other areas with different layouts and sizes of irrigated lands, and three years of rainfall and solar time series. These tests found consistent key findings with the case study presented.

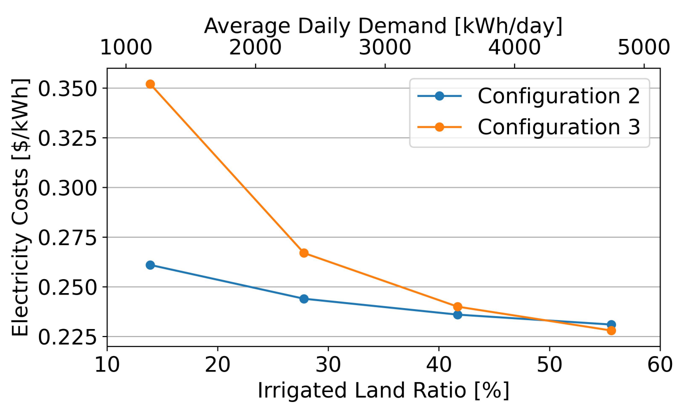


Fig. 4. Changes in electricity total costs with increasing irrigated land ratio, comparing the irrigation zonal connected Configuration 2 and the whole region connected Configuration 3. The bottom x-axis shows the ratio of irrigated land to the total land area of all irrigation zones, while the top x-axis shows the average daily load following irrigation expansion.

## IV. Conclusion and Recommendations

Electricity systems can improve crop production and local livelihoods in rural regions with irrigation demand. This paper builds on previous research on irrigation land detection. For regions with detected clusters of irrigation lands, we investigate different electricity system solutions, including standalone systems, mini-grid options, and grid-connected options, considering three key characteristics of irrigation loads: flexibility, seasonality, and minimum pump power requirements. These different energy systems are simulated and compared. Additionally, we studied how the costs of mini-grid systems evolve with increased irrigation lands.

Recommendations for policymakers and stakeholders: (a) Develop electricity systems for clusters of smallholder farms that are already engaging in irrigation but not yet using electricity, due to their clear energy demands. The combination of the irrigation detection algorithm and the energy system design model we developed can effectively identify and serve these clusters; (b) Enhance daytime solar energy utilization by leveraging the flexibility of irrigation loads; (c) Shared solar systems between farmers can enhance solar utilization and solve the oversizing issue from the minimum pump power requirement; (d) Adopt small-scale, low-voltage mini-grids (Configuration 2) for optimal energy utilization and cost efficiency under current conditions. Start with these small systems; as irrigation areas and demand grow, these systems can be upgraded to larger regional medium-voltage connected systems (Configuration 3).

This study primarily focuses on one case, although some sensitivity tests were conducted in the same region. Future work will involve testing the models in various regions to validate their adaptability and improve the models' flexibility and scalability for simulating different solar-irrigation projects.

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